Ritwik Sarkar

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Key skills

- Machine Learning
- Computer Vision
- Data Modeling
- Natural Language Processing
- Tensorflow
- generative ai
- LLM
- Google Cloud Platforms
- Deep Learning
- Data Analysis
- Machine Learning Algorithms
- Pytorch
- Data Preprocessing
- Continuous Learning
- Transformers
- LangChain
- Big Data

Personal Information

City **Durgapur**

Country **INDIA**

Languages

- English

Profile Summary

- **Machine Learning Engineer** dedicated to leveraging cutting-edge technologies to solve real-world problems efficiently.
- **Strong technical foundation** coupled with a **business analytical mindset,** facilitating effective collaboration with teams to achieve strategic objectives.
- **Proven expertise** in the **end-to-end lifecycle development** of machine learning models, from data collection and preprocessing to deployment and monitoring in production environments.
- **Experience in areas of** TensorFlow, PyTorch, Scikit-Learn, Keras, Hugging Face Transformers and Fine Tuning, with strong proficiency in Python, SQL, data wrangling, and deep learning frameworks.
- **Proficient in using cloud platforms** such as **AWS and GCP** for scalable machine learning and AI solutions.
- **Additional knowledge in Proficient in state-of-the-art methodologies** such as LLAMA, OpenAI, Langchain, Gemma, Pinecone, etc. and advanced techniques including vector databases and knowledge graphs. Committed to delivering robust and scalable AI solutions.

Education

B.A, 2018

Durgapur Government College, Bardhaman

12th, 2015

West Bengal, Bengali / Bangla

Social links

https://github.com/ritwiks9635

10th, 2013

West Bengal, Bengali / Bangla

Projects

5 Days

Named Entity Recognition

- **Objective:** Developed a Named Entity Recognition (NER) model to identify and classify entities (e.g., people, organizations, locations) in text, enhancing text analysis capabilities.
- **Technologies Used:** Python, NLTK, SpaCy, TensorFlow
- **Methodology:**
  - Implemented a deep learning model using LSTM and CRF layers to recognize named entities in unstructured text.
  - Fine-tuned a pre-trained BERT model to improve accuracy in entity recognition across different domains.
- **Impact:** Achieved an F1-score of 0.90, enabling more accurate extraction of key information from large datasets, which facilitated better insights in natural language processing tasks.

3 Days

Online Payment Fraud Detection

- **Objective:** Developed a machine learning model to detect fraudulent transactions in online payment systems.
- **Technologies:** Python, Scikit-Learn, Pandas, Numpy, Matplotlib
- **Highlights:** Implemented and optimized models like Random Forest and Gradient Boosting, achieving 95% accuracy. Utilized SMOTE to handle data imbalance, significantly improving detection rates.
- **Impact:** Deployed model reduced false positives by 15%, enhancing the efficiency of fraud detection processes.

2 Days

Age and Gender Detection

- **Objective:** Developed a deep learning model to predict the age and gender of individuals from facial images.
- **Technologies Used:** Python, TensorFlow, Keras, OpenCV, Numpy, Matplotlib
- **Methodology:**
  - Used a convolutional neural network (CNN) architecture for feature extraction from facial images.
  - The model was trained on a labeled dataset with diverse age

groups and genders, achieving high accuracy in predictions.

- **Impact:** The model was able to accurately predict age and gender with an accuracy of 94%, making it suitable for applications in demographic analysis and personalized marketing.

2 Days

Traffic Signs Detection

- **Objective:** Developed a computer vision model to detect and classify traffic signs in real-time, enhancing the safety and efficiency of autonomous driving systems.
- **Technologies Used:** Python, OpenCV, TensorFlow, Keras
- **Methodology:**
  - **Data Preparation:** Collected and labeled a dataset of traffic signs. Preprocessed images through resizing, normalization, and data augmentation.
  - **Model Architecture:** Implemented a Convolutional Neural Network (CNN) for feature extraction and classification of traffic signs.
  - **Performance:** Achieved a classification accuracy of 98% on the test set, with real-time detection capabilities.
- **Impact:** The model improved traffic sign recognition in autonomous vehicles, contributing to safer navigation in complex road environments.

3 Days

Density Estimation

- **Objective:** Developed a model to estimate the probability distribution of data points in high-dimensional space using Gaussian Mixture Models (GMMs).
- **Technologies Used:** Python, Scikit-Learn, NumPy, Tensorflow, Matplotlib, Seaborn
- **Methodology:**
  - Implemented GMM to model the data distribution, allowing for flexible density estimation with multiple Gaussian components.
  - Applied Expectation-Maximization (EM) algorithm for parameter estimation, optimizing the likelihood of the data under the model.
- **Impact:** Successfully identified underlying data patterns, enabling more accurate predictions in anomaly detection and clustering tasks.

3 Days

Face Mask Detection

- **Objective:** Developed a real-time face mask detection system to identify whether individuals are wearing masks, aimed at enhancing public health safety during the COVID-19 pandemic.
- **Technologies Used:** Python, TensorFlow, OpenCV, Keras, Matplotlib, numpy
- **Methodology:**
  - Preprocessed a dataset of images labeled as "mask" and "no mask."
  - Trained a Convolutional Neural Network (CNN) model to classify images based on the presence of a mask.
  - Integrated the model with OpenCV for real-time detection through webcam feeds.
- **Impact:** Achieved an accuracy of 98%, allowing for reliable mask detection in public spaces, with potential applications in automated entry systems and monitoring tools.

2 Days

Brian Tumor Detection uaing YOLO V7

- **Objective:** Developed a real-time brain tumor detection system using the YOLOv7 (You Only Look Once) object detection model to accurately identify tumors in MRI scans.
- **Technologies Used:** Python, TensorFlow, YOLOv7, OpenCV, Matplotlib, numpy
- **Methodology:**
  - **Data Preparation:** Collected and labeled a dataset of MRI brain images with annotations for tumor regions.
  - **Model Implementation:** Trained the YOLOv7 model to detect and localize tumors in MRI images with high precision.
  - **Evaluation:** Achieved a detection accuracy of 93% with minimal false positives.
- **Impact:** Enabled faster and more accurate tumor detection, aiding in early diagnosis and treatment planning.

1 Days

Next Word Prediction

- **Objective:** Developed a machine learning model to predict the next word in a sequence of text, enhancing typing efficiency and autocomplete functionalities.
- **Technologies Used:** Python, TensorFlow, Keras, LSTM, NLTK, Numpy
- **Methodology:**
  - Preprocessed text data by tokenizing and creating sequences.
  - Trained a Long Short-Term Memory (LSTM) model to learn

context from previous words and predict the most likely next word.
- **Impact:** Achieved an accuracy of 90% on test data, significantly improving text input speed and accuracy in applications.

2 Days

Text Generator

- **Objective:** Developed a text generation model to generate coherent and contextually relevant text based on a given prompt.
- **Technologies Used:** Python, Hugging Face Transformers, GPT-2
- **Methodology:** Fine-tuned a pre-trained GPT-2 model on a custom dataset to improve text generation capabilities. Implemented the model using the Hugging Face Transformers library and optimized it for generating human-like text.
- **Impact:** The model was able to generate paragraphs of text that maintained logical flow and context, demonstrating potential for content creation and conversational AI applications.

3 Days

One shots learning for Face Verification

- **Objective:** Developed a face verification system using one-shot learning to accurately identify individuals based on a single example image.
- **Technologies Used:** Python, Numpy, TensorFlow, Keras, Siamese Network, Deep Learning
- **Methodology:**
  - Implemented a Siamese Network that learns to distinguish between pairs of images by comparing their feature embeddings.
  - Trained the model using a contrastive loss function to minimize the distance between embeddings of the same person and maximize the distance for different people.
- **Impact:** Achieved a 98% accuracy in face verification with minimal data, making the system highly effective in scenarios where only one image per person is available.

2 Days

Chatbot using Transformer

- **Objective:** Developed an intelligent chatbot capable of engaging in natural language conversations with users, leveraging advanced Transformer models.
- **Technologies Used:** Python, Hugging Face Transformers, PyTorch
- **Methodology:**

    - **Model Selection:** Fine-tuned a pre-trained GPT-2 model from Hugging Face for conversational AI tasks.
    - **Training:** Trained the model on a large dataset of dialogues to generate contextually relevant and coherent responses.
    - **Integration:** Deployed the chatbot in a web application, enabling real-time user interaction.
- **Impact:** The chatbot demonstrated a 90% accuracy in providing meaningful responses, significantly enhancing user engagement and satisfaction.

Certification

- Sequence Model
- Google Cloud Bigdata And Machine learning
- How Google does Machine Learning
- Structuring Machine learning Project
- Improving Deep Neural Networks: Hyperparameter Tuning, Regularization and Optimization
- Introduction to TensorFlow for Artificial Intelligence, Machine Learning, and Deep Learning
- Machine Learning for All
- Launching into Machine Learning
- Mathematics For Machine Learning Linear Algebra
- Supervised Machine Learning: Regression, Classification
- Advanced Learning Algorithm
- Mathematics For Machine learning: Multivariate Calculus
- Convolutional Neural Networks
- Unsupervised Learning, Recommenders, and Reinforcement Learning
- Mathematic For Machine learning: PCA